Branch Coverage Analysis Report

Lab 6 – Code and Branch Coverage | Ahana H Sharan | PES1UG24AM020

1. Objective & Method

Check that every decision in food_delivery.py is evaluated both True and False, and raise branch coverage to at least 90%.

2. Coverage Progression

Test file	Tests	Missed stmts	Partial branches	Branch Coverage	Missing
test_food_delivery.py (starter)	2	21	5	30%	6, 8, 11-14, 16, 23-29, 33-40
CodeCoverageTest_food_delivery_application.py	11	0	3	95%	34→36, 36→40, 38→40
BranchCoverageTest_food_delivery_application.py	13	0	0	100%	–

After the code coverage tests, every line ran but 3 branches in estimated_time were still untested, because its only test used is_peak_hour=True, restaurant_busy=True.

3. Uncovered Branches and Test Cases Added

Missing branch	Meaning	Test added	Input -> Expected
34→36	is_peak_hour is False	test_not_peak_not_busy	estimated_time(5, False, False) → 35
36→40	restaurant_busy is False	test_not_peak_not_busy	(same test)
38→40	busy, but nested is_peak_hour is False	test_busy_not_peak	estimated_time(5, False, True) → 50

4. Result

All **13 tests passed**. The result is **24/24 branches** taken, 0 partial branches and 0 missed statements, which gives **100% branch coverage** (target ≥ 90%).

```
============================================== 13 passed in 0.08s ==============================================
PS C:\Projects\SWE\L6_L7_20_food_delivery\Lab 6> python -m coverage report -m
Name                                    Stmts   Miss Branch BrPart  Cover   Missing
-----------------------------------------------------------------------------------
BranchCoverageTest_food_delivery.py        31      0      0      0   100%
food_delivery.py                           33      0     24      0   100%
-----------------------------------------------------------------------------------
TOTAL                                      64      0     24      0   100%
```

5. Analysis

• Branch coverage is stricter than line coverage. The suite already had 100% line coverage, but branch coverage showed 95% because the False outcomes of the decisions in estimated_time were never run.

• The nested if is_peak_hour (line 38) needed its own test. Here, only the busy and not-peak case reaches its False branch.

• Limitation: coverage.py measures branches, not individual conditions. In distance_km <= 0 or distance_km > 15, only the <= 0 side is exercised, and WELCOME with a first order is never tested. Both still count as 100% branch coverage.